

Demonstration of Proposed Features

Date	3 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status	Demonstrated (Yes/No)	Remarks
1	Crop Recommendation	Predicts best crop using N, P, K, temp, humidity, pH, rainfall via Logistic Regression	Implemented	Yes	94% accuracy achieved
2	Data Analysis	Univariate, Bivariate & Multivariate analysis using Matplotlib and Seaborn	Implemented	Yes	All plots generated in notebook
3	Data Preprocessing	Null value check, outlier removal (IQR), seasonal crop extraction, train/test split	Implemented	Yes	Clean dataset ready for ML
4	K-Means Clustering	Groups similar agricultural conditions using Elbow method (4 clusters identified)	Implemented	Yes	Elbow graph plotted
5	Flask Web Application	3-page web app: Home, About, FindYourCrop with navigation and prediction form	Implemented	Yes	Runs on localhost:5000
6	Real-time Prediction	User enters parameters → Flask backend → ML model → Crop recommendation displayed instantly	Implemented	Yes	Tested with multiple inputs

Feature Implementation Summary:

Metric	Count / Value
Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%